RESECON 303- Honors Colloquium

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**Farming Under Capitalism: A Comparative Study of Agriculture Systems in China and the United States**

In the contemporary globalised economy, the food products found in the ordinary grocery store are the culmination of complex systems of production. Products like cereal, packaged meats, and condiments are ordinary products that seem simple to the ordinary consumer. However, the production of these products is much more complex than the ordinary consumer may be aware. This paper starts by challenging the simplicity of the food products found in the ordinary grocery store. It seeks to shed light on the systems of production of these food products. It will discuss key issues like the ownership of farms and agricultural resources, the determinants of the cost of basic commodities like seeds, the patent system, and the role of corporations in the food industry. It is the intention of the paper to offer a comprehensive overview of the systems of production of food products.

I'm from India, where farming is something most people have a personal connection to, such as a grandparent's land, a village market, rice grown by your own family. Coming to the U.S., I expected the food system here to be just as humane, just a much bigger version of it. What I found was something different. A system that is extraordinarily efficient in some ways, but also deeply concentrated and controlled by a surprisingly small number of corporations that most people have never heard of. And then, through this class and reading books like Blockchain Chicken Farm, I began to learn about China's version of the same issue. Two very different

nations, two very different systems of government, but in both cases, the farming communities are being pressured by forces beyond their control.

This paper will explore how, from seed patents and meatpacker monopolies in America to blockchain and state-led investments in rural China, as well as the meeting points between both systems, such as when a Chinese company acquires America's biggest pork supplier and when African American communities from North Carolina live in proximity to its pig farms. What ties everything together is the idea that, whether or not there is state intervention in capitalism, as is the case in China and America, respectively, farmers will always be left behind. Wendong Zhang, an agricultural economist at Iowa State who grew up farming in rural China, puts it, “the well-being of farm households and the farm sector are of perennial significance in China and the United States”.[1]

Let's take something you probably do every two weeks: walking down the cereal aisle. There are a lot of boxes in those aisles. But if you consider the actual producers of all these packages, you will find that there are only a few of them. The same holds true for the next stage down the food chain, where we have seeds and livestock. A 2023 USDA report found that just two seed companies controlled 72 per cent of planted corn acres and 66 per cent of planted soybean acres in the United States.[2] Two companies. That's almost all the corn going into your Doritos, your Sprite, and the chicken feed that became your McDonald's nuggets.

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[1]Wendong Zhang, "The Commonalities and Differences between Chinese and U.S. Agriculture," Agricultural Policy Review (Fall 2015), Iowa State University Center for Agricultural and Rural Development, https://agpolicyreview.card.iastate.edu/fall-2015/commonalities-and-differences-between-chinese-and-us-agriculture.

[2]James M. MacDonald, Xiao Dong, and Keith O. Fuglie, Concentration and Competition in U.S. Agribusiness, U.S. Department of Agriculture, Economic Research Service, EIB-256 (June 2023), p. iii.

This kind of market structure, where a few firms run an entire industry, is what economists call concentration. And it's not just seeds. By 2019, four meatpacking companies handled 85 per cent of steer and heifer slaughter nationally.[3] Well, the next time you pick up a package of ground beef at your local grocery store, there is a pretty good chance that it went through one of these four companies on its way there. This is not really what a free market is supposed to be all about. Such firms have immense influence over the price the farmer can fetch from his livestock.

How did this happen? There is no better example than that of the seeds industry. Before the eighties, many crop varieties were developed by the public universities, and farmers could freely use the varieties. But then came biotechnology. Farmers soon learned that they could add specific genes to a plant, and create varieties that could resist pests, such as bugs in corn, or be immune to herbicides, such as soybeans. The modified plants quickly caught on. In 2020, it is estimated that 55 percent of all land under cultivation in the US used genetically modified seeds[4]. The ketchup you spread on your burger? It probably came from farms where GM soybeans were grown in rotation with tomatoes. The corn syrup in nearly every soda you've ever had. Almost certainly GMO.

With science came patents. Companies could now legally own seed varieties and charge farmers to use them and they couldn't save seeds from one season to plant the next. Between 1990 and 2020, seed prices rose by 270 percent across the board. For GMO crops specifically, the jump was 463 per cent. [5] Farmers who had been in the habit of re-planting saved seeds, like their parents before them, suddenly found themselves tied into a yearly contract with Bayer, Corteva, or Syngenta. Consolidation in the industry was felt most intensely between 2015 and

[3]MacDonald, Dong, and Fuglie, p. iii.
[4] MacDonald, Dong, and Fuglie, p. 7 per cent,.
[5]MacDonald, Dong, and Fuglie, p. 15.

2018, when a string of mergers reduced the number of major global seed producers from six to four. Meatpacking was another industry in which consolidation was a major issue. JBS, Tyson, Cargill, and National Beef controlled the industry. Because cows can only travel so far before slaughter, farmers in much of the country may only have one or two buyers. A study found that farmers in highly concentrated areas received 9 per cent less for their cows than farmers in less concentrated areas.[6] That's a significant cut, year after year, with no negotiating power to push back. None of this shows up clearly on food labels. But it shows up in rural America in declining small farms, in communities that used to be built around independent feed stores and local cooperatives and are now just stops on corporate supply chains. And it shows up in grocery bills that have kept climbing faster than wages.

Now picture something totally different. A village in a remote province in China, unpaved roads, chickens wandering around, and an older farmer checking on her flock. That's the world Xiaowei Wang takes you into in *Blockchain Chicken Farm*. But here's the thing that makes it strange and fascinating: those chickens have GPS trackers on their legs. Every time a chicken moves, eats, or gets processed, that data goes onto a blockchain, the same underlying technology behind Bitcoin and crypto.[7]

At first, it may seem absurd. Why is it even necessary for the farmers on a remote farm to use blockchain technology? However, Wang clarifies it all. Consumers from affluent cities such as Shanghai and Beijing can justifiably distrust the products they consume. Within the last few years, China experienced a crisis where its milk formula contained melamine. There have been documented cases of restaurant oil recycled from drainage ditches, which people started calling

[6]MacDonald, Dong, and Fuglie, pp. 25–31.
[7]Xiaowei Wang, Blockchain Chicken Farm: And Other Stories of Tech in China's Countryside (FSG Originals, 2020), pp. 1–30.

gutter oil. After those kinds of scandals, urban shoppers will pay significantly more for food they can trace. Blockchain gives them a QR code on your chicken package, and you can see exactly which farm it came from, what it ate, and how it lived. Think of it as a very high-tech version of a farm-to-table receipt.

That technology didn't get out to rural villages on its own. The Chinese government helped push it there as part of larger poverty alleviation and rural modernisation programs, partnering with companies like Alibaba and JD.com to connect farmers to urban buyers through smartphone platforms.[8] This is what state capitalism in agriculture actually looks like: the farms are privately run, but the government shapes where investment flows, builds the infrastructure to make e-commerce possible in poor areas, and sets national goals that private companies are expected to help meet.

In the U.S., agricultural technology is also advancing. John Deere now sells self-driving tractors, and AI is being used to analyse satellite crop data. But the difference is who gets access. Large operations with capital can afford it; smaller farms generally can't. The Chinese approach, imperfect as it is, at least tries to get technology to farmers who would otherwise be left out of it entirely.

Wang doesn't romanticise any of this, though. She's clear-eyed about what's happening. The blockchain system is designed primarily to serve wealthy city consumers, not the farmers using it. The farmers are still earning modest incomes. They become dependent on platforms and systems they don't own and can't fully control. The data flowing off their farms feeds into corporate and government databases, which they have no real say in. Wang writes about visiting grandmothers who have farmed the same land for decades, now being expected to navigate

[8]Wang, Blockchain Chicken Farm, pp. 45–70.

smartphone apps just to sell their eggs. The technology is there, but the power that benefits who controls it, is still flowing away from the village.[9]

There's also a real gap between China's ambitions and where things currently stand. A 2024 research study comparing China and the U.S. on agricultural modernisation found that China's farming sector today is roughly equivalent to where America was back in the 1960s or 70s, about 60 years behind.[10] One American agricultural worker can support about 144 people. In China, that figure is 8.[11] Part of that reflects that land, a typical U.S. farm is 400 to 500 acres.[12] A typical Chinese farm is under 2 acres.[13] The scale difference alone makes the comparison complicated.

[9]Wang, pp. 80–110.
[10]Yuying Yang, Chuanbo Chen, and Yiying Sun, "Challenges in Chinese Path to Modernization of Agriculture and Rural Areas: A Comparison with the U.S.," Journal of Agricultural Science 16, no. 8 (2024): 15–30, doi:10.5539/jas. v16n8p15, p. 15.
[11]Yang, Chen, and Sun, p. 25.
[12] Zhang, "Commonalities and Differences."
[13]Zhang, "Commonalities and Differences."

If you go into a Kroger and you want to buy Smithfield bacon, it is a Chinese-owned company. In 2013, a Chinese company named WH Group, which is affiliated with the Chinese government, acquired Smithfield Foods, the largest pork producer in the United States, for 4.7 billion dollars.[14] It was the largest Chinese acquisition of an American company at the time. The reason behind it, basically, was that China is worried about feeding itself.

China has roughly the same land mass as the U.S. but more than four times the population, and it only holds about 7 to 9 per cent of the world's arable land.[15] That land is also shrinking. It has been reported by the Chinese Government that about 19 percent of China's total agricultural land was polluted due to heavy industry and years of rapid growth without considering environmental conservation.[16] Meanwhile, as more Chinese people moved into the middle class, they started eating more meat. More meat means more grain for animal feed. In 2021 alone, China imported 28.35 million metric tons of corn, 152 per cent more than the year before.[17]

Buying Smithfield was a hedge against all that uncertainty by creating a pork reserve. It gave China immediate access to established farms, processing plants, cold storage, and distribution networks it couldn't have built in less than decades. When the African Swine Fever tore through China between 2018 and 2020, killing hundreds of millions of pigs and cutting national pork output by nearly half, Smithfield's American operations kept exporting to China and helped stabilize prices on the other side of the world.[18]

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[14]Lauren Greenwood, China's Interests in U.S. Agriculture: Augmenting Food Security through Investment Abroad, U.S.-China Economic and Security Review Commission Staff Research Report (May 26, 2022), pp. 11–12.
[15]Greenwood, p. 5.
[16]Greenwood, p. 5.
[17]Greenwood, p. 3.
[18]Greenwood, pp. 11–12.

The communities surrounding the hog farms operated by Smithfield in Eastern North Carolina have been suffering the effects for years. The waste of thousands of hogs are kept in large open lagoons, which not only emit hazardous gaseous compounds but also spill their contents into local water bodies during extreme weather conditions[19]. According to a study done in 2018, people living in close proximity to these operations suffer from high incidences of sepsis, kidney diseases, tuberculosis, and infant mortality, as opposed to those living farther away[20]. This is not merely an issue of figures, these are children getting asthmatic and parents consuming poisonous water, all at the expense of deteriorating community health and well-being without any solution in sight. Nor are these affected communities at random. Most often than not, these facilities locate themselves next to impoverished and minority black communities in rural America. Communities that had undergone generations of economic and social marginalization, which reduced their ability to confront the growth of this industry around them. And China takes advantage of the environmental degradation its pork producers leave for American minorities to cope with.

China's agricultural footprint in the U.S. goes beyond Smithfield. Chinese companies have purchased grain elevators in Ohio, taken stakes in grain trading firms like Lansing Trade Group, and gained joint access to river terminals used to export corn and soybeans.[21] And China has also gone after agricultural technology through less legitimate means. There are documented cases of Chinese researchers stealing GM corn seeds from test fields in Iowa, and of a Monsanto employee stealing a proprietary algorithm about soil nutrition and attempting to bring it back to

[19] Greenwood, pp. 14.
[20] Greenwood, pp. 14.
[21] Greenwood, pp. 13.

China.[22] A single hybrid seed can cost $30 to $40 million in research costs to develop. Stealing it saves China years of work.

So which system works better? It is hard to overstate agricultural productivity in America, and both fail in ways that are hard to defend. The U.S. leads the world in exports of soybeans, corn, beef, and pork. Yields per acre have climbed steadily for decades. But that productivity comes with costs that don't show up on the price tag at the grocery store. Concentrating the seed industry into a few patent-holding corporations has made farming more expensive for the farmers themselves.[23] Concentrating meatpacking into four major players has suppressed the prices ranchers get for their livestock.[24] The environmental costs of nitrogen runoff into rivers, soil compaction from industrial tillage, and methane from factory farms are real and massive and mostly unaccounted for in market prices.

And it's not just production. Food retail has consolidated dramatically, too. Back in 1990, the four biggest grocery chains controlled 13 per cent of national food sales. By 2019, that number had risen to 34 per cent. [25] In rural counties, the picture is even bleaker: studies show that rural grocery stores are highly concentrated and, in fact, monopolies in rural America. If you're from a small town, you probably know what I'm talking about: one store, prices you cannot argue with, and nowhere else to go.

China's system is very different from the outside. The government builds rural roads, subsidises irrigation, and sets minimum purchase prices for key crops like rice and wheat so

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[22]Greenwood, pp. 15–16.
[23]MacDonald, Dong, and Fuglie, p. 7.
[24]MacDonald, Dong, and Fuglie, pp. 25–31.
[25]MacDonald, Dong, and Fuglie, p. 36.

farmers don't get crushed by market crashes. Programs like the blockchain chicken initiative, whatever their limitations, did create new income options for some of the country's poorest rural households.[26] These are things private markets don't do on their own; private capital goes where profit is, not where need is.

But Chinese farmers still face disadvantages that the government programs haven't fixed. The average government farm payment in China is $113 per farm. In the U.S., it's $9,925.[27] Rural Chinese residents earn just 54 per cent of the national average income, and the urban-rural income gap in China has barely moved since 1978.[28] Farmers selling through platforms like Taobao are generating data and value that mostly flows to the platform company, not back to the village. The form of exploitation changes, but the direction doesn't.

If you've eaten anything with corn syrup, soybean oil, or canola oil ingredients found in most packaged snacks, fast food, salad dressings, and cooking oils at American grocery stores, chances are, you've consumed a genetically modified ingredient. Over 90 per cent of the corn, soybeans, canola, and sugar beets grown in the U.S. are genetically modified.[29] The FDA assures that these GMO foods are safe to eat, and most scientists agree. But the lingering question ultimately isn't about safety; it's about power. Concentration of GMO seed ownership in two or three companies means farmers pay dramatically higher prices, have fewer choices, and can't do something as basic as save seed from their own crop. Their grandparents could. They can't.

In China, the GMO politics are kind of fascinating and contradictory. Many Chinese consumers are deeply sceptical of genetically modified food; there have been public protests, and

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[26]Wang, Blockchain Chicken Farm, pp. 80–110.
[27]Zhang, "Commonalities and Differences."
[28]Yang, Chen, and Sun, pp. 24–25.
[29]MacDonald, Dong, and Fuglie, p. 7.

for years, Chinese officials publicly opposed it. But China is simultaneously one of the world's biggest importers of GMO soybeans and corn, mostly to feed animals. A lot of the tofu, pork, and chicken eaten across China was raised on GMO soybeans grown in Iowa or Brazil. The whole blockchain chicken thing is partly a reaction to this tension: consumers who don't trust GM-fed meat are willing to pay extra to know exactly what went into the animal they're eating.[30]

These two food systems are completely tangled up with each other. One in every four rows of soybeans planted in Iowa is exported to China.[31] A Chinese company owns America's biggest pork producer. Chinese researchers have stolen American seed technology. American agribusiness corporations sell GMO seeds and chemicals worldwide, including China. You cannot understand what is on your plate in America or China without understanding how these two systems interact.

[30]Greenwood, p. 3.
[31]Zhang, "Commonalities and Differences.colour"

As I reflect on the contents of two distinctly different shopping carts, one from America's supermarkets and one from China's wet markets, I can't help but think of how they represent two distinctly different cultures. America's supermarket shopping cart is stocked with heavily processed foods, snack foods, and inexpensive meats. Plenty of them, cheap in calorie count, yet terribly unhealthy. On the other hand, China's wet market shopping cart is increasingly full of fresh produce, whole grains, and protein sources that we can actually trace to their source.

America is a world leader in exporting agricultural products. America has large-scale farms, advanced technology, and companies with very high profits. However, this level of production capacity has not provided security to the average Americans who do not have access to food. From what I have seen after visiting Trader Joes and Target myself, fresh foods such as fruits are priced higher than processed foods; a bag of chips is cheaper than a bag of apples. Also, a homemade meal using fresh vegetables ends up being pricier than fast food, for example if I order Popeyes on Doordash there is a buy one get one offer on chicken wings for 8 dollars. The United States Department of Agriculture (USDA) has reported that about 18.3 million Americans were food insecure in 2024.[32] This means that they lack affordable nutritious food necessary for a healthy living. The wealth created by agriculture in America trickles up to the companies and shareholders rather than the farmers.[33] This leaves an American farmer between the vice of being squeezed by the big meatpacking firms and paying costly seeds.

The Chinese system seems worse on paper. Its farms have an average size of less than 2 acres compared to the average farm size of 400 to 500 acres in America.[34] The Chinese farmer

[32] USDA Economic Research Service, Food Security Data-2024
[33] [3] MacDonald, Dong, and Fuglie, pp. iii, 25–31.

[34] Zhang, "Commonalities and Differences."

makes only 54 percent of the national average income level.[35] However, the extensive government engagement in food production has achieved one thing: the American system cannot boast of a sincere dedication to the task of making sure that people actually eat. China uses price floors for rice and wheat products, ensuring availability of basic foods at affordable prices despite market volatility. Rural development and the use of consumer-oriented websites by the Chinese authorities shorten the distance between the farmer and the consumer[36]. The "Blockchain Chicken" project described by Wang demonstrates how China approaches issues related to food safety as an important public issue, rather than a luxury for people who can afford it.[37] Despite all their flaws, consumers in China today can get a fresher and more transparent product than Americans, simply because their system is designed to provide such a product.[38]

This is not to say that China's approach is superior to America's. The rural-urban income difference is no lower today than it was in 1978.[39] Chinese farmers are still at the mercy of technologies that are not their own.[40] The state considers feeding China its top priority, and the welfare of its farmers its secondary objective.[41] However, it would be unfair to assume that America is the only country with an effective economic policy while China is simply an inefficient and unsuccessful state-owned enterprise. America has its agribusiness sector, and China has a state that, at least, makes sure its people have something to eat.

For the U.S., however, the solution must begin with disentangling corporate monopoly power in the areas of seeds, meatpacking, and food retailing. From 1990 to 2020, the price of

[35] Yang, Chen, and Sun, "Challenges in Chinese Path," pp. 11
[36] Greenwood, pp. 9–10.
[37] Wang, pp. 80–110.
[38] Wang, pp. 45–70.
[39] Yang, Chen, and Sun, p. 11.
[40] Wang, pp. 100–130.
[41] Greenwood, p. 8.

genetically modified seed increased by 463% as a result of increased consolidation.[42] Public funding in agriculture through research institutions such as the traditional land-grant colleges, which used to be free until the advent of agricultural science privatization, can contribute to leveling the playing field.[43] In the case of China, bridging the investment gap between the government in the area of food systems and the income received by farmers becomes critical. The disposable income of rural residents is currently only 54% of the national average.[44]

The ability of China to leverage its strengths through the involvement of rural communities in developing new platforms and technologies becomes essential.

[42] MacDonald, Dong, and Fuglie, p. 6 and p.21.
[43] Zhang, "Commonalities and Differences."
[44] Yang, Chen, and Sun, pp. 11.

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